

## ch9 Momentum

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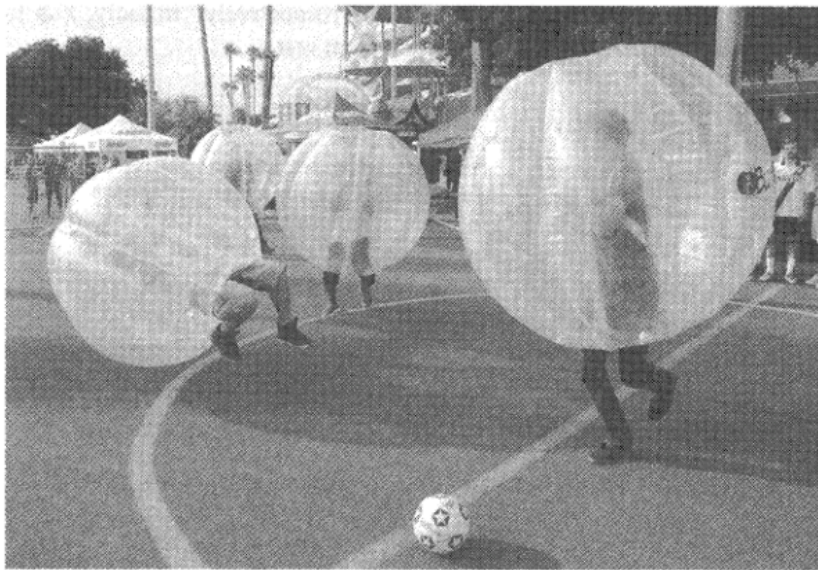
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2017 Q12

12. Players of “bubble soccer” wear air-filled plastic “bubbles” as shown.



Which of the following statements best explains why the bubble can reduce the chance of injury during a collision ?

- A. The bubble increases the mass of the player, thus the momentum of the player increases.
- B. The bubble increases the air resistance acting on the player.
- C. The bubble lengthens the impact time during a collision.
- D. Like a balloon, the bubble provides a lifting force to the player.

2022 Q7

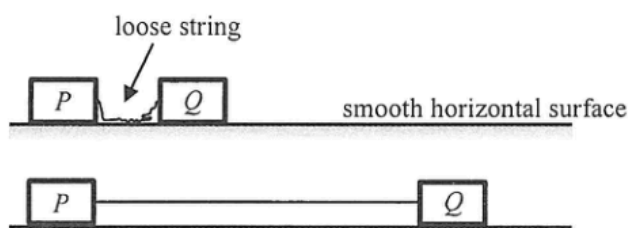
7. An egg is released from a height and falls towards a cushion without breaking.



Which of the following is the most probable explanation ?

- A. The cushion makes the egg stop at a shorter distance.
- B. The cushion helps lengthen the time of impact.
- C. The cushion helps reduce part of the gravitational force acting on the egg.
- D. The cushion increases the reaction force acting on the egg during impact.

8. On a smooth horizontal surface, two identical blocks  $P$  and  $Q$  are connected by a light inextensible string. The blocks are at rest and the string is loose initially.

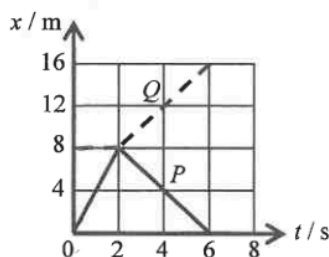
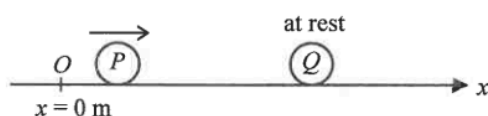


$Q$  is given a speed of  $4 \text{ m s}^{-1}$  and moves to the right. Find the speeds of the blocks when the string just becomes taut and  $P$  starts to move.

|    | block $P$            | block $Q$            |
|----|----------------------|----------------------|
| A. | $1 \text{ m s}^{-1}$ | $1 \text{ m s}^{-1}$ |
| B. | $2 \text{ m s}^{-1}$ | $1 \text{ m s}^{-1}$ |
| C. | $2 \text{ m s}^{-1}$ | $2 \text{ m s}^{-1}$ |
| D. | $4 \text{ m s}^{-1}$ | $2 \text{ m s}^{-1}$ |

## 2022 Q12

12.



On a smooth horizontal surface, sphere  $P$  travelling along the  $x$ -axis collides head-on with another sphere  $Q$ , which is initially at rest at  $x = 8 \text{ m}$ . The graph shows the displacement-time ( $x-t$ ) relationship of  $P$  (solid line) and  $Q$  (dotted line). The collision takes place at  $t = 2 \text{ s}$  and the collision time is negligible. Which statement is correct?

- The collision is perfectly inelastic.
- The mass of  $P$  is larger than that of  $Q$ .
- $P$  still moves along  $+x$  direction after collision.
- $P$  and  $Q$  move with the same speed after collision.

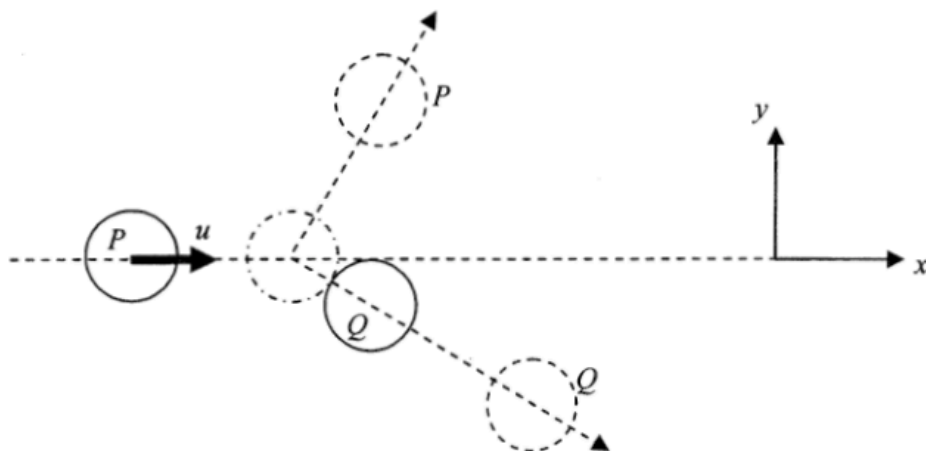
## 2018 Q12

12. At a certain instant, an object flying horizontally to the right at  $1 \text{ m s}^{-1}$  suddenly explodes into two fragments of mass ratio  $1 : 2$ . If just after the explosion the more massive fragment flies to the right at  $3 \text{ m s}^{-1}$ , what would happen to the other fragment just after the explosion?

- It would fly to the left at  $3 \text{ m s}^{-1}$ .
- It would fly to the left at  $4 \text{ m s}^{-1}$ .
- It would be momentarily at rest.
- It would fly to the right at  $1 \text{ m s}^{-1}$ .

## 2012 Q13

13. On a smooth horizontal surface, a circular disk  $P$  moving at velocity  $u$  along the  $x$  direction collides obliquely with an identical disk  $Q$  initially at rest as shown below. The mass of each disk is  $m$ . Which statements about the collision is/are correct?



- (1) Momentum of the system in the  $y$  direction is not conserved.
  - (2) The total kinetic energy of  $P$  and  $Q$  after collision is  $\frac{1}{2}mu^2$  if the collision is perfectly elastic.
  - (3) Speed of  $Q$  after collision is less than  $u$ .
- A. (1) only  
 B. (3) only  
 C. (1) and (2) only  
 D. (2) and (3) only

## 2013 Q10

10.

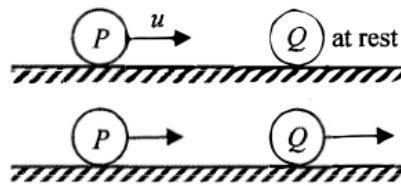


Two identical spheres are moving in opposite directions with speeds  $u$  and  $v$  (with  $u > v$ ) respectively as shown. They make a head-on collision. Which of the following diagrams show(s) a *possible* situation of the spheres after collision?

- (1)
- (2)
- (3)

- A. (1) only  
 B. (3) only  
 C. (1) and (2) only  
 D. (2) and (3) only

12. On a smooth horizontal surface, a marble  $P$  moving with speed  $u$  collides head-on with another marble  $Q$ , which is at rest. After collision,  $P$  and  $Q$  move with different speeds as shown.

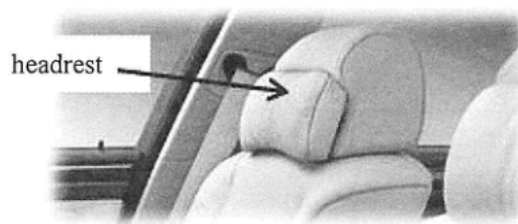


Which of the following statements about this collision is/are correct ?

- (1) During collision, the force acting on  $Q$  by  $P$  is equal and opposite to that acting on  $P$  by  $Q$ .
- (2) The total momentum of the two marbles is conserved only when the collision is perfectly elastic.
- (3) The kinetic energy lost by  $P$  must be equal to that gained by  $Q$ .

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

5.



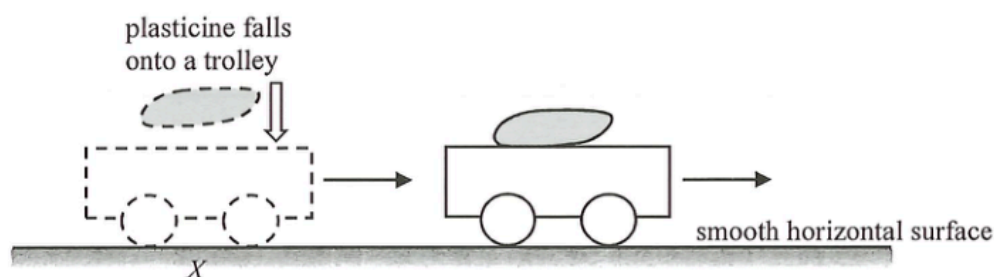
For a car travelling on a highway, which of the following statements about the safety design of the headrest is/are correct ?

- (1) As the headrest is soft, it can reduce the force exerted on the passenger's head during impact.
- (2) It can minimise injury of the passenger when the car is struck by another one from behind.
- (3) It can minimise injury of the passenger when the car brakes suddenly.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

## 2023 Q11

11. A trolley moves with a constant velocity along a smooth horizontal surface. When the trolley reaches point  $X$ , a plasticine falls vertically onto it. After the collision, they stick together and continue to move forward.

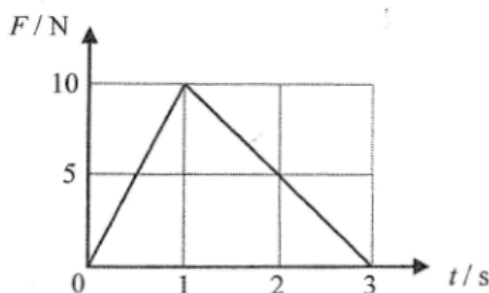


Which description about the **total linear momentum** of the trolley and the plasticine just before and just after collision is correct?

|    | along the horizontal direction | along the vertical direction |
|----|--------------------------------|------------------------------|
| A. | conserved                      | conserved                    |
| B. | conserved                      | not conserved                |
| C. | not conserved                  | conserved                    |
| D. | not conserved                  | not conserved                |

## 2015 Q6

6.

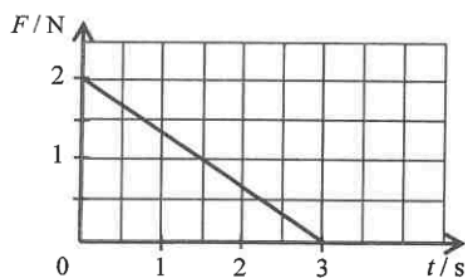


An object of mass 3 kg is initially at rest on a smooth horizontal ground. A force  $F$  is applied horizontally to the object such that the magnitude of  $F$  varies with time  $t$  as shown. What is the speed of the object at  $t = 3$  s? Neglect air resistance.

- A.  $2.5 \text{ m s}^{-1}$   
 B.  $5 \text{ m s}^{-1}$   
 C.  $10 \text{ m s}^{-1}$   
 D.  $15 \text{ m s}^{-1}$

## 2022 Q8

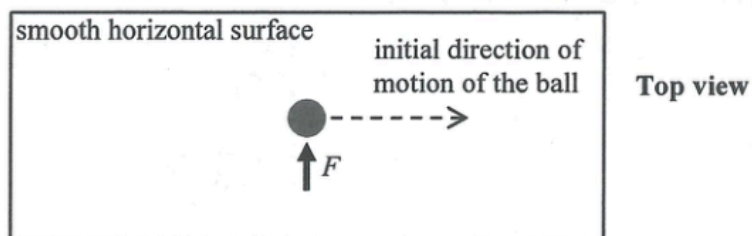
8. An object of mass  $2\text{ kg}$  moves with velocity  $1\text{ m s}^{-1}$  initially. It is acted upon by a force  $F$  which varies with time  $t$  as shown in the graph. The force and the object's velocity are in the same direction. Find the speed of the object at  $t = 3\text{ s}$ .



- A.  $1.5\text{ m s}^{-1}$   
 B.  $2.5\text{ m s}^{-1}$   
 C.  $3.5\text{ m s}^{-1}$   
 D.  $4.5\text{ m s}^{-1}$

## 2019 Q10

10.



The above figure shows a ball moving with a constant speed along a straight line on a smooth horizontal surface. At a certain instant, the ball is acted on by a force  $F$  for a very short time as shown above. Which **subsequent** path below would the ball most closely follow?

A.



B.



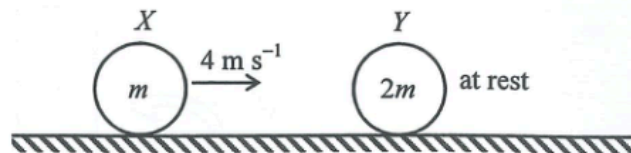
C.



D.



11.



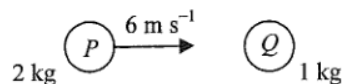
On a smooth horizontal surface, sphere  $X$  of mass  $m$  travels with speed  $4 \text{ m s}^{-1}$ . It collides head-on with another sphere  $Y$  of mass  $2m$ , which is at rest initially. Which of the following can be the speed of  $Y$  just after collision?

- (1)  $1 \text{ m s}^{-1}$                       (2)  $2 \text{ m s}^{-1}$                       (3)  $3 \text{ m s}^{-1}$

- A. (1) only  
 B. (2) only  
 C. (1) and (2) only  
 D. (2) and (3) only

2014 Q7

7.



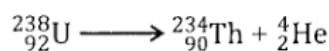
A sphere  $P$  of mass  $2 \text{ kg}$  makes a head-on collision with another sphere  $Q$  of mass  $1 \text{ kg}$  which is initially at rest. The speed of  $P$  just before collision is  $6 \text{ m s}^{-1}$ . If the two spheres move in the same direction after collision, which of the following could be the speed(s) of  $Q$  just after collision?

- (1)  $2 \text{ m s}^{-1}$   
 (2)  $4 \text{ m s}^{-1}$   
 (3)  $6 \text{ m s}^{-1}$

- A. (1) only  
 B. (1) and (2) only  
 C. (2) and (3) only  
 D. (1), (2) and (3)

2021 Q8

8. A stationary uranium nucleus  ${}^{238}_{92}\text{U}$  decays to give a thorium nucleus  ${}^{234}_{90}\text{Th}$  and an  $\alpha$  particle  ${}^4_2\text{He}$ .



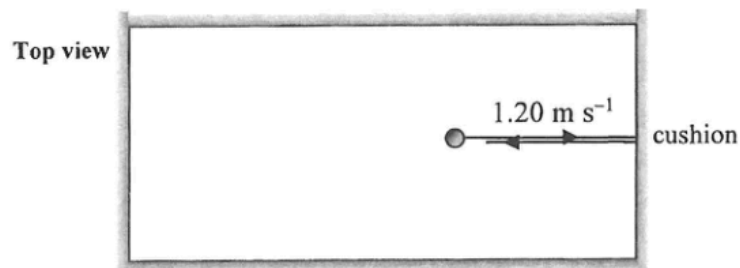
Which of the following correctly describes the situation about the  ${}^{234}_{90}\text{Th}$  nucleus and the  $\alpha$  particle just after the decay?

- |    | magnitude of momentum $p$  | kinetic energy KE                          |
|----|----------------------------|--------------------------------------------|
| A. | $p(\text{Th}) = p(\alpha)$ | $\text{KE}(\text{Th}) < \text{KE}(\alpha)$ |
| B. | $p(\text{Th}) > p(\alpha)$ | $\text{KE}(\text{Th}) > \text{KE}(\alpha)$ |
| C. | $p(\text{Th}) = p(\alpha)$ | $\text{KE}(\text{Th}) > \text{KE}(\alpha)$ |
| D. | $p(\text{Th}) = p(\alpha)$ | $\text{KE}(\text{Th}) = \text{KE}(\alpha)$ |



## 2024 Q11

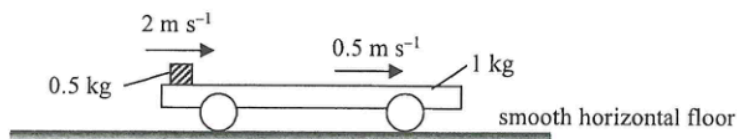
11. On a billiard table, a billiard ball of mass  $0.16 \text{ kg}$  and moving with a uniform speed of  $1.20 \text{ m s}^{-1}$  collides elastically with the cushion. It rebounds with the same speed normal to the cushion as shown. If the impact time is  $0.0003 \text{ s}$ , find the average force acting on the ball by the cushion during collision.



- A.  $0 \text{ N}$   
 B.  $640 \text{ N}$   
 C.  $1280 \text{ N}$   
 D.  $2560 \text{ N}$

## 2026 Q10

10. A  $1\text{-kg}$  trolley with a  $0.5\text{-kg}$  block placed on its horizontal platform is resting on a smooth horizontal floor. The block is then given a sharp push to the right. At an instant after the push, the block and the trolley are moving to the right at  $2 \text{ m s}^{-1}$  and  $0.5 \text{ m s}^{-1}$  respectively as observed from the floor.

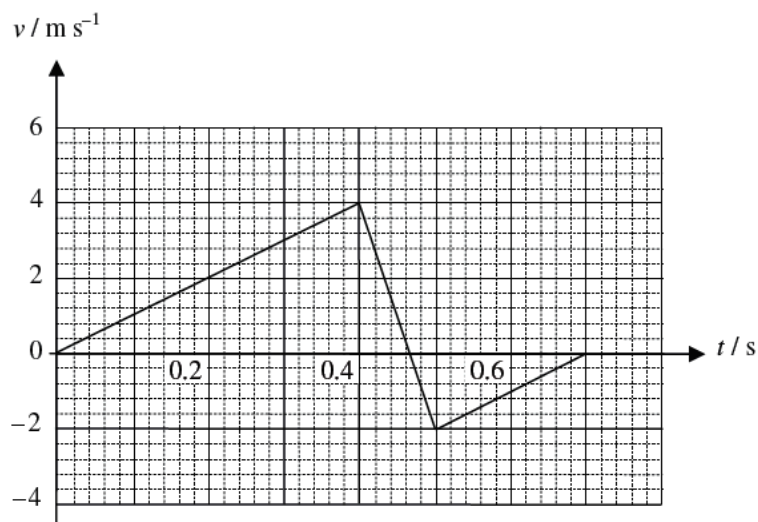


Finally, they travel together with the same speed  $v$ . Find  $v$ . Neglect air resistance.

- A.  $0.5 \text{ m s}^{-1}$   
 B.  $1 \text{ m s}^{-1}$   
 C.  $1.5 \text{ m s}^{-1}$   
 D.  $2 \text{ m s}^{-1}$

pp Q13

13.



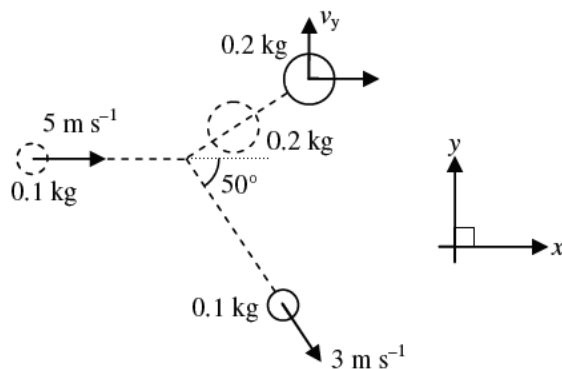
A ball of mass  $0.2 \text{ kg}$  is released from rest. It hits the ground and rebounds. The velocity-time graph of the ball is shown above. Which of the following statements are correct?

- (1) The magnitude of the change in momentum of the ball during the collision is  $1.2 \text{ kg m s}^{-1}$ .
- (2) The magnitude of the average force acting on the ball by the ground during the collision is  $12 \text{ N}$ .
- (3) There is mechanical energy loss during the collision.

- A. (1) and (2) only  
 B. (1) and (3) only  
 C. (2) and (3) only  
 D. (1), (2) and (3)

pp Q14

- \*14. A disc of mass  $0.1 \text{ kg}$  and velocity  $5 \text{ m s}^{-1}$  strikes a stationary disc of mass  $0.2 \text{ kg}$  on a smooth table. After the collision, the  $0.1 \text{ kg}$  disc moves with a speed of  $3 \text{ m s}^{-1}$  at  $50^\circ$  to the  $x$  direction. Find the component of the velocity of the  $0.2 \text{ kg}$  disc in  $y$  direction,  $v_y$ , after the collision.



- A.  $1.15 \text{ m s}^{-1}$   
 B.  $1.54 \text{ m s}^{-1}$   
 C.  $1.92 \text{ m s}^{-1}$   
 D.  $2.01 \text{ m s}^{-1}$

sap Q13

13. A car  $P$  of mass 1000 kg moves with a speed of  $20 \text{ m s}^{-1}$  and makes a head-on collision with a car  $Q$  of mass 1500 kg, which was moving with a speed of  $10 \text{ m s}^{-1}$  in the opposite direction before the collision. The two cars stick together after the collision. Find their common velocity immediately after the collision.

- A.  $2 \text{ m s}^{-1}$  along the original direction of  $P$
- B.  $2 \text{ m s}^{-1}$  along the original direction of  $Q$
- C.  $14 \text{ m s}^{-1}$  along the original direction of  $P$
- D.  $14 \text{ m s}^{-1}$  along the original direction of  $Q$